

FORM TP 2013015



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JANUARY 2013

CARIBBEAN EXAMINATIONS COUNCIL

**CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION**

INFORMATION TECHNOLOGY

Paper 032 – General Proficiency

ALTERNATIVE TO SBA

1 hour 30 minutes

17 JANUARY 2013 (a.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of FOUR questions.
2. Answer ALL questions.
3. Number EACH answer correctly in the answer booklet.

**Ministry of Education
Sports and Youth Affairs**

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01229032/JANUARY 2013

Answer ALL questions.

1. WORD PROCESSING

The Election Authority of a CARICOM state is responsible for General Elections. Three parties are contesting the general election to be held in November 2013. Computers and related equipment will be used to prepare documents, spreadsheets, databases and to develop computer programs to be used to handle the various tasks related to the elections.

The following letter is sent to the leaders of the three parties contesting the election.

THE ELECTION AUTHORITY
Office of the Director

11 May 2013

Mr John Persaud
Leader of Unity Party

Dear Mr Persaud:

National Election 2013

The Prime Minister has announced that the General Election will be held on 20 November 2013. As such, please note that Nomination Day will be 20 September 2013. Your complete list of candidates must be submitted to the election officials between 2:00 p.m. to 5:00 p.m. at the City Hall. Failure to submit your complete list of fifty candidates will result in your party not being able to contest the election.

Patrick Collins
Director

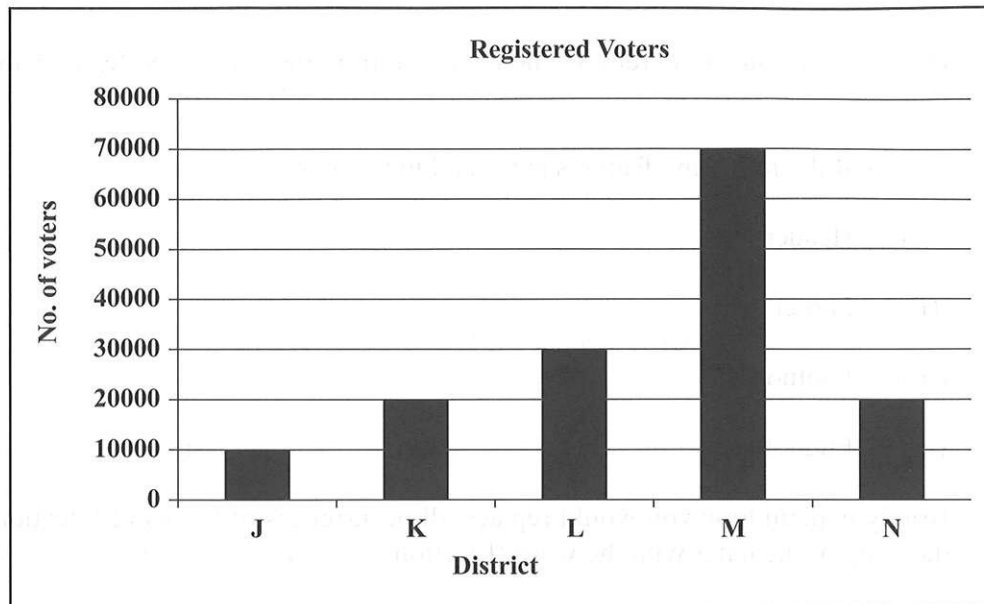
Ensuring Free and Fair Elections

- (a) State the line spacing and justification that have been applied to the text in the body of the letter. **(2 marks)**
- (b) State THREE formatting features that have been used for emphasis in the letter. **(3 marks)**
- (c) Outline how you would remove the underline from the date of the Nomination Day. **(2 marks)**
- (d) Which of the following features are found in the letter?
 - (i) Header
 - (ii) Footer
 - (iii) Footnote
 - (iv) Endnote**(2 marks)**
- (e) Briefly explain how you would replace all occurrences of the word 'election' ONLY in the body of the letter with the word 'Election'. **(4 marks)**
- (f) The election authority has recently designed a logo. Explain how you would insert the logo into the letter above the text "THE ELECTION AUTHORITY". **(4 marks)**
- (g) A mail merge was used to prepare the letter on page 2. State the name of the document that contains the merge fields. **(1 mark)**
- (h) State the number and names of merge fields that this letter should have. **(2 marks)**

Total 20 marks

2. SPREADSHEET

Consider the following chart which was created from the data in a spreadsheet.



- (a) Name the type of chart shown. (1 mark)
- (b) Write the name of another type of chart which would be suitable to represent the data in the spreadsheet. (1 mark)
- (c) State the title of the chart and the title of the horizontal axis. (2 marks)
- (d) What is the purpose of the chart? (1 mark)

- (e) Using the chart on page 4, copy and complete the spreadsheet below which was used to create the chart.

1	District	No. of Voters
2		
3		
4		
5		
6		

(5 marks)

- (f) Using the spreadsheet completed in Part (e),
- (i) write a FUNCTION based on the data to calculate the TOTAL number of registered voters (2 marks)
 - (ii) state an appropriate format for the values entered in the No. of Voters column in Part (e). (1 mark)
- (g) The data in the spreadsheet were sorted in DESCENDING order by the No. of Voters column.
- (i) Write the RANGE OF CELLS used to perform the sort. (1 mark)
 - (ii) Which district would be at the top of the sorted list? (1 mark)
- (h) You wish to filter the data for all districts with over 25,000 registered voters. Copy and complete the table below in your answer booklet with the criterion that would filter the data.

District	Voters

(2 marks)

- (i) Write the results produced from Part (h). (2 marks)
- (j) If the = sign is omitted from the function **max(B2..B6)** in cell B10, what would appear in the cell? (1 mark)

Total 20 marks

3. DATABASE MANAGEMENT

Consider the following database containing two tables:

TABLE 1: DISTRICT

DCODE	DNAME
A	BCE
B	ESI
C	ESC
D	DEM
E	INT

TABLE 2: POLLING

DCODE	PTYPE	PLACES
D	Government	200
E	Private	20
A	Government	120
C	Private	25
B	Government	90
D	Private	50
A	Private	30
C	Government	75
E	Government	60
B	Private	50

(a) State

- (i) the number of records and fields in the POLLING table (2 marks)
- (ii) the appropriate data types for the PLACES and DCODE fields in the POLLING table (2 marks)
- (iii) appropriate field(s) in the POLLING table that can be considered a primary key (2 marks)
- (iv) the name of the field used to join the two tables. (1 mark)

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- (b) If the data in the POLLING table (Table 2) is arranged in ascending order by the PLACES field, state the PTYPE that would be at the top of the table. **(1 mark)**

- (c) The result of a query is shown below:

SumOfPLACES	
	720

- (i) What does the number 720 represent? **(1 mark)**
- (ii) State the name of the table that was used in this query. **(1 mark)**
- (d) The following query was created on the data in the database:

Field:	DNAME	TYPE	PLACES
Table:	DISTRICT	POLLING	POLLING
Sort:			
Show:	☒	☒	☒
Criteria:		'Government'	>100
or:			

- (i) Explain the purpose of the query. **(1 mark)**
- (ii) State the name(s) of the table(s) used to create the query. **(1 mark)**
- (iii) State number of records that will be displayed when the query is executed. **(1 mark)**
- (e) The following query was created on the data in the database:

Field:	PLACES	cost: [PLACES]*10
Table:	POLLING	
Sort:		
Show:	☒	☒
Criteria:		
or:		

- (i) State the type of field that is created called **cost**. **(1 mark)**
- (ii) Name the type of data that would be found in the **cost** field. **(1 mark)**

(f) The following report was created.

ELECTION POLLING PLACES		
DNAME	TYPE	PLACES
BCE		
	Government	120
	Private	30
Summary for 'DNAME' = BCE (2 detail records)		
Sum		150
DEM		
	Government	200
	Private	50
Summary for 'DNAME' = DEM (2 detail records)		
Sum		250
ESC		
	Government	75
	Private	25
Summary for 'DNAME' = ESC (2 detail records)		
Sum		100
ESI		
	Government	90
	Private	50
Summary for 'DNAME' = ESI (2 detail records)		
Sum		140
INT		
	Government	60
	Private	20
Summary for 'DNAME' = INT (2 detail records)		
Sum		80
Grand Total		

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Name

- (i) the field used to group the records in the report (1 mark)
- (ii) the field(s) used to sort the records in the report (2 marks)
- (iii) ONE summary feature used in the report (1 mark)
- (iv) the title of the report. (1 mark)

Total 20 marks

4. PROGRAMMING AND PROBLEM SOLVING

Three parties registered to contest an election. One of the parties later withdrew, therefore only two of them obtained votes. The following program segment refers to the election results obtained.

(a) The declaration statements are listed below.

Line No.	Program Code
1	Program Winner;
2	
3	Const base = 100;
4	
5	Var
6	party1, party2: integer; {votes obtained by the two parties}
7	lg: integer;
8	percent: real;

- (i) State TWO data types used in the program and indicate the line number(s) where the data types are used. **(4 marks)**
- (ii) Explain the statement in line 3. **(4 marks)**
- (iii) (a) What type of error would occur if the semicolon (;) was missing at the end of line 8? **(1 mark)**
(b) State the phase of the program when this error would occur. **(1 mark)**
- (iv) Rewrite the code in line 6 if all three parties had contested the election. **(2 marks)**

(b) The body of the program is as shown below.

Line No.	Program Code
9	Begin
10	lg := 0;
11	Write ('Enter the votes obtained by the two parties: ');
12	Readln (party1, party2);
13	If (party1 > party2) then
14	lg := party1
15	Else
16	lg := party2;
17	pct := (lg/(party1+ party2)) * base;
18	Writeln ('The winning party has',pct, ' percent of
19	votes cast');
20	End.

- (i) What does the variable **lg** represent in the program and what is its initial value? **(3 marks)**
- (ii) State the type of control structure used in the program. Indicate the line numbers where this control structure is used. **(2 marks)**
- (iii) State the line number for an arithmetic operation used in the program. **(1 mark)**
- (iv) Copy and complete the following trace table if the numbers 10,000 and 30,000 were entered for party1 and party2 respectively.

p1	p2	lg	pct

(5 marks)

- (c) What will be the output of the program? **(1 mark)**
- (d) Draw a flowchart to represent lines 12–16 of the PASCAL program. **(6 marks)**

Total 30 marks

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END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.